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Unpacking conflicting evaluations and ambivalence in online hotel booking: The moderating role of perceived enjoyment in user retention

Yixiu Yu [a,*], Fred Davis [b], Eric Walden [b], Ofir Turel [c]

[a] Miller College of Business, Ball State University, Muncie, IN, 47306, USA
[b] Rawls College of Business, Texas Tech University, Lubbock, TX, 79409, USA
[c] School of Computing and Information Systems, University of Melbourne, Parkville, Victoria, 3010, Australia

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ABSTRACT

The online travel industry, particularly hotel booking platforms, operates in a highly competitive environment where user retention is crucial for reducing acquisition costs and ensuring long-term profitability. While ambivalence toward technology has been linked to unfavorable outcomes, its antecedents and associations with user retention in the context of online hotel booking platforms remain underexplored. This study addresses this gap by examining how key platform features—perceived security risk, effort, usefulness, and enjoyment—are associated with user ambivalence and retention. Using structural equation modeling with survey data from 387 participants, the findings reveal that perceived usefulness and enjoyment are negatively associated with ambivalence, whereas perceived effort and security risk are positively associated with it. Ambivalence, in turn, is negatively associated with user retention. Additionally, perceived enjoyment moderates this relationship by weakening the negative association between ambivalence and retention. Mediation analyses suggest that ambivalence partially mediates the relationships between perceived usefulness and enjoyment with retention, while fully mediating the associations between perceived effort and security risk with retention. These findings highlight the importance of addressing user ambivalence and provide actionable insights for enhancing user retention. Strategies such as improving enjoyable features and addressing perceived effort and security concerns may contribute to increased engagement and competitiveness in online hotel booking platforms.

1. Introduction

The online travel industry has experienced substantial growth, with hotel booking platforms playing a pivotal role in facilitating this expansion (Manes and Tchetchik, 2018). Platforms such as Expedia, Booking.com, and TripAdvisor provide users a wide array of options and features that streamline travel planning and booking processes (Kim et al., 2021; Teresa Borges-Tiago et al., 2021). Despite these benefits, the industry's highly competitive nature and low switching costs makes user retention critical for reducing platforms' acquisition expenses and ensuring long-term profitability (Ullah et al., 2019). Consequently, understanding the factors associated with user retention has become a priority for researchers and practitioners seeking to enhance platform performance and user loyalty.

Recent Information Systems (IS) research has begun to explore ambivalence—conflicting feelings or evaluations toward information systems—and its relevance to user behavior (Qahri-Saremi and Turel, 2020; Turel and Qahri-Saremi, 2023; Yu and Davis, 2024). Ambivalence arises when users experience both positive and negative reactions to a system, creating uncertainty in their evaluations and decisions. Prior studies have primarily focused on ambivalence in contexts such as social media (Qahri-Saremi and Turel, 2020) and online dating platforms (Chen et al., 2023), where social and relational dynamics often drive these conflicting evaluations. For example, social media users frequently report discomfort stemming from ambivalence and adopt various coping strategies to manage mixed feelings (Qahri-Saremi and Turel, 2020). Similarly, in online dating platforms, ambivalence is shaped by both positive and negative social influences, which is negatively associated with users' intentions to adopt these systems (Chen et al., 2023).

However, the role of ambivalence in online hotel booking platforms remains underexplored. This gap is significant because online hotel booking platforms exhibit unique features and interaction dynamics compared to other domains. Unlike social media, where ambivalence may stem from privacy concerns or social pressures, and online dating

* Corresponding author.
E-mail address: yyu4@bsu.edu (Y. Yu).

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platforms, where emotional risks are a dominant factor, online hotel booking platforms present distinct challenges. These include high financial stakes (Morosan and DeFranco, 2016), strict decision-making timelines (Guo and Li, 2022), substantial risks associated with failed plans (San-Martín et al., 2020), and the cognitive effort required to process extensive options and information (Xiang et al., 2015). Additionally, users must often share sensitive personal and financial data during the booking process, intensifying security concerns (Talwar et al., 2020). These challenges, coupled with time-sensitive decisions and low switching costs, increase the complexity of user experiences and may heighten ambivalence, potentially influencing outcomes such as retention and platform switching.

Exploring ambivalence and its triggers in online hotel booking is critical for gaining a balanced understanding of user experiences and informing actionable interventions. Ambivalence reflects a complex evaluative process where users simultaneously recognize a platform's strengths and weaknesses, making it an important indicator of both satisfaction and dissatisfaction (Yu and Davis, 2024). If left unaddressed, ambivalence can result in user disengagement, platform abandonment, or migration to competitors. Investigating its antecedents and associations within this context contributes to theoretical advancements while offering practical insights for improving platform design and user loyalty.

To address these gaps, this study introduces a research model examining the relationships among core platform features, ambivalence, and user retention in online hotel booking. Specifically, it investigates how perceived security risk, effort, usefulness, and enjoyment are associated with user ambivalence and how ambivalence, in turn, relates to user retention. Additionally, this study explores whether positive user experiences, such as perceived enjoyment, can moderate the negative association between ambivalence and user retention. By unpacking these relationships, this study advances the understanding of ambivalence and offers actionable strategies for designing online booking platforms that promote user retention in a competitive marketplace.

2. Theoretical background

2.1. Ambivalence theory

The term *ambivalence* originates from the Latin *ambo* (both) and *valere* (to be strong), referring to the experience of two opposing forces (Thompson et al., 1995). In psychology, ambivalence is broadly conceptualized as a state of psychological conflict arising from the simultaneous presence of opposing evaluations, emotions, or beliefs (Priester and Petty, 1996; Thompson et al., 1995; Van Harreveld et al., 2015). Unlike indifference, which reflects an absence of strong feelings, ambivalence involves the coexistence of conflicting yet strong feelings (Thompson et al., 1995). For example, individuals may experience ambivalence when watching the film *Life is beautiful*, simultaneously feeling happiness and sadness, or when graduating from college, feeling both joy and nostalgia (Larsen et al., 2001). This duality has been shown to influence decision-making processes and has been extensively explored in psychological research (Conner and Armitage, 2008; Schneider and Mattes, 2021).

Ambivalence is driven by specific triggers—situational factors or conditions that create tension, dilemmas, or contradictions within a context (Ashforth et al., 2014). These triggers generate oppositional forces (e.g., a demand for A and not-A) that manifest in four major forms: (1) hybrid identities, contradictory goals, and role conflicts; (2) dualities; (3) multifaceted objects; and (4) temporal factors. These frameworks provide insights into the formation of ambivalence in consumer and user experiences. For example, luxury shoppers may encounter ambivalence stemming from hybrid identities as they balance aspirational self-images (luxury seekers) with practical or financial responsibilities (Prestini and Sebastiani, 2021). In wedding planning, ambivalence may emerge from role conflicts, such as reconciling personal preferences with family expectations (Otnes et al., 1997). During the COVID-19 pandemic, ambivalence among patrons of luxury restaurants reflected dualities, where emotional attachment conflicted with heightened risk perceptions (Peng and Chen, 2021). Similarly, green hotel guests often experience ambivalence from multifaceted evaluations, feeling positive about eco-friendly programs like education and innovation while having negative views of practices such as water conservation or green purchasing (Yu et al., 2017). Ambivalence also evolves through a dynamic interplay of psychological and contextual factors. For example, in the case of luxury restaurant reservations during the pandemic (Peng and Chen, 2021), ambivalence may diminish as pandemic-related risks decreased and perceived control increased. This demonstrates how situational and psychological factors can interact, shaping the intensity of ambivalence over time (Ashforth et al., 2014).

Building on foundational work, ambivalence is now recognized as a distinct construct with meaningful implications for decision-making in complex or high-stakes situations. Unlike related concepts such as indecisiveness or dissatisfaction, ambivalence emphasizes the simultaneous coexistence of positive and negative evaluations (Priester and Petty, 1996; Van Harreveld et al., 2015). In digital environments, this duality is particularly relevant as users navigate complex trade-offs involving usability, security, and enjoyment.

2.2. Ambivalence in online hotel booking

In the context of online hotel booking, user ambivalence refers to the coexistence of conflicting evaluations—whether in attitudes, emotions, or beliefs—toward a booking platform, resulting in cognitive tension and decision-making challenges. This conflict often arises from contrasting evaluations of platform features. For example, users may value a platform's usefulness and convenience but simultaneously express concerns about security risks, the cognitive effort required to navigate the platform, or the reliability of user-generated content. These dynamics align with the ambivalence triggers identified by Ashforth et al. (2014), where opposing roles, expectations, and experiences contribute to psychological tension.

The formation of ambivalence in online hotel booking reflects the interplay between instrumental goals (e.g., efficiency, economic value) and emotional experiences (e.g., enjoyment, trust). Users aim to secure the best deals and maximize economic value while seeking a seamless and trustworthy user experience. However, when platforms present conflicting priorities, functional tensions may arise, contributing to ambivalence. Examples of these tensions include:

Effort vs. Usefulness: Users often appreciate robust search filters and detailed reviews but may feel frustrated by the high cognitive effort required to navigate complex interfaces or process overwhelming amounts of information. This tension reflects the challenge of reconciling instrumental benefits with usability burdens.

Enjoyment vs. Security Concerns: Personalized recommendations and visually appealing interfaces can enhance perceived enjoyment and emotional attachment to the platform. However, these same features may raise concerns about data privacy and ambiguous platform policies (Akhtar et al., 2022). Regulatory frameworks such as the General Data Protection Regulation (GDPR) have increased user awareness of online risks (Talwar et al., 2020). Although such regulations are designed to foster trust, they may paradoxically increase ambivalence by simultaneously reassuring users of safeguards and amplifying concerns about data misuse.

Multi-Dimensional Evaluations: Platforms often evoke ambivalence through contradictory evaluations. For example, users may trust a hotel booking platform's reliability but remain concerned about data security. Similarly, emotional attachment to features like personalized recommendations may coexist with frustrations over usability issues, such as hidden fees, restrictive cancellation policies, or ambiguous terms. These inconsistencies result in conflicting assessments of the platform's overall value.

Temporal Factors: Urgency cues (e.g., "Only 2 rooms left!") can create tension between the perceived need to act quickly and the desire for autonomy in decision-making. As trip dates approach, user priorities often shift from thorough research to convenience and availability, intensifying the conflict between efficiency and emotional satisfaction.

These examples illustrate how online hotel booking platforms act as dual-faceted objects, simultaneously eliciting positive evaluations (e.g., usefulness, enjoyment) and negative evaluations (e.g., security risks, high effort). The coexistence of these conflicting evaluations generates cognitive tension, which may influence users' decision-making processes and retention behaviors.

Recognizing these instrumental-emotional dualities and temporal dynamics provides valuable opportunities for platforms to address ambivalence, reduce cognitive tension, and enhance user retention. This study builds on these observations to examine the psychological mechanisms underlying ambivalence formation and its associations with user retention in online hotel booking platforms.

2.3. IS literature on user ambivalence

The integration of ambivalence into IS research is expanding as scholars increasingly recognize its relevance in shaping user experiences across various systems. User often exhibit ambivalence and mixed feelings toward information systems, weighing their perceived benefits against drawbacks (Qahri-Saremi and Turel, 2020; Turel and Qahri-Saremi, 2023; Walden et al., 2005; Yu and Davis, 2024). Despite its importance, IS research on ambivalence is still in its early stages, presenting substantial opportunities for further exploration (Lakhiwal et al., 2023; Qahri-Saremi and Turel, 2020; Turel and Qahri-Saremi, 2023; Yu and Davis, 2024).

Existing studies have highlighted how ambivalence may be linked to distinct coping strategies—either flexible (adaptive) or inflexible (avoidance)—influenced by individual personality traits and contextual factors (Qahri-Saremi and Turel, 2020). For example, users with high neuroticism may adopt rigid coping mechanisms, amplifying ambivalence's influence on their behavior. In cybersecurity contexts, ambivalence has been linked to protection motivation and coping appraisal (Ng et al., 2021). Similarly, in online purchasing, ambivalence is associated with attention and decision-making processes (Lakhiwal et al., 2023). Table 1 summarizes key findings and contexts explored in existing IS literature on user ambivalence.

Despite these advancements, notable gaps persist in the literature. First, most studies focus on ambivalence in technology adoption contexts (e.g., Chen et al., 2023), leaving its role in post-adoption behaviors, such as platform retention, underexplored. Second, some studies address ambivalence implicitly through its consequences without explicitly defining or measuring it as a standalone construct (e.g., Stein et al., 2015). Third, while ambivalence has been studied in contexts like social networking sites (Qahri-Saremi and Turel, 2020; Turel and Qahri-Saremi, 2023) and online dating platforms (Chen et al., 2023), the unique environment of online hotel booking—characterized by high financial stakes, cognitive demands, and security concerns—have yet to be thoroughly investigated. This study aims to address these gaps by developing and testing a research model that examines how perceptions of platform-specific core features—security risk, effort, usefulness, and enjoyment—are associated with ambivalence and user retention. It also explores whether perceived enjoyment moderates the relationship between ambivalence and retention.

Table 1
Summary of existing IS journal articles on user ambivalence.

Literature	Journal	IS Context	Key findings
Chen et al. (2023)	*Information Technology & People*	Online dating platforms	Positive and negative social influence shapes ambivalence, which subsequently affects adoption intention.
Dang and Liu (2021)	*Computers in Human Behavior*	AI robots	Robot mind is positively associated with ambivalence.
Lakhiwal et al. (2023)	*MIS Quarterly*	E-commerce websites	Ambivalence influences purchase intention.
Moody et al. (2014)	*Electronic Commerce Research and Applications*	E-commerce websites	When trust and distrust co-exist, user ambivalence is activated.
Moody et al. (2017)	*European Journal of Information Systems*	E-commerce websites	Revealed the non-linear complex relationships between trust, distrust, and ambivalence.
Ng et al. (2021)	*Journal of Management Information Systems*	Cybersecurity	Ambivalence influences protection motivation and coping appraisal.
Qahri-Saremi and Turel (2020)	*Journal of Management Information Systems*	Social network sites	Users' coping responses to ambivalence are shaped by their neuroticism levels, affecting subsequent post-adoption behaviors.
Turel and Qahri-Saremi (2023)	*Information Systems Journal*	Social network sites	Identified various coping mechanisms individuals employ when confronted with conflicting feelings.
Stein et al. (2015)	*MIS Quarterly*	Office software	Mixed emotions are common in IT use.
Yu and Davis (2024)	*Journal of Computer Information Systems*	Office software, social networking sites, e-commerce websites, gaming apps, and online banking	Satisfaction and dissatisfaction are distinct dimensions and can co-exist in ambivalent situations.

2.4. HCI literature on user experience

User experience (UX) research in Human-Computer Interaction (HCI) extends beyond traditional IS research by emphasizing user perceptions through hedonic, temporal, contextual, and aesthetic values in interactive technology use (Hassenzahl and Tractinsky, 2006). Widely abbreviated as UX, user experience refers to users' emotional responses and overall perceptions when interacting with technology (Hassenzahl, 2008; Hassenzahl and Tractinsky, 2006). This research stream explores both positive and negative experiences (Tuch et al., 2013), investigates various UX dimensions (Yu et al., 2024), and provides comprehensive frameworks for evaluating systems, such as e-commerce platforms (Son and Kim, 2023) and travel booking platforms (Wani et al., 2017).

Norman's (1988) seminal work on user-centered design principles highlights the importance of designing platforms that align with users' needs and cognitive abilities. These principles focus on usability and intuitive design, which directly influence perceived effort. Building on this foundation, Hassenzahl (2010) introduced experience-oriented design and hedonic quality, emphasizing the holistic, emotional, and psychological aspects of interacting with a product. This approach prioritizes factors such as pleasure, engagement, and meaning to enhance user satisfaction (Hassenzahl, 2010; Hassenzahl and Tractinsky, 2006). Subsequent studies have shown that perceived enjoyment often has a stronger association with user attitudes than perceived usefulness or ease of use (Hornbæk and Hertzum, 2017). For example, research on hedonic systems has found that positive emotions associated with perceived enjoyment significantly can increase user satisfaction (Pereira and Tam, 2021). In emerging technologies, perceived enjoyment has

been identified as a key predictor of user attitudes, often surpassing perceived usefulness in driving the adoption of AR wearables (Holdack et al., 2022).

These insights are critical for understanding how online booking platform features—particularly user perceptions of enjoyment—interact with ambivalence and behavioral intentions. Bilgihan et al. (2015) applied flow theory to hotel booking platforms research, demonstrating that immersive and enjoyable experiences are associated with increased user engagement and loyalty. Incorporating perceived enjoyment into platform design can help practitioners identify competitive advantages and enhance website stickiness (Wani et al., 2017). Delivering high-quality, positive user experiences while minimizing negative ones is essential for reducing ambivalence and ensuring the platform's long-term success.

3. Research model and hypotheses

3.1. Research model

As shown in Fig. 1, this study proposes a research model to examine the mechanisms underlying user ambivalence in online hotel booking. Specifically, we investigate the relationships among user ambivalence, user retention, and key antecedent beliefs, including perceived security risk, effort, usefulness, and enjoyment. These variables were selected because they are well-established in technology adoption and user experience literature and are highly relevant to the context of online hotel booking platforms. They represent key dimensions influencing user behavior and decision-making (Franque et al., 2021; Yan et al., 2021). Specifically, perceived security risk is widely acknowledged in technology adoption literature as a key factor negatively associated with trust and satisfaction, especially in online transaction contexts (Featherman and Pavlou, 2003; Lee, 2009). Perceived effort reflects usability challenges and aligns with the Technology Acceptance Model (TAM) (Davis, 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003), where ease of use plays a central role in technology adoption. Perceived usefulness, a core construct of TAM (Davis, 1989), represents the utilitarian value of a system, which is essential for understanding user retention in task-oriented platforms. Perceived enjoyment, as a hedonic factor (Davis et al., 1992), captures the emotional and experiential aspects of user engagement, complementing utilitarian dimensions (Hassenzahl and Tractinsky, 2006). Online booking platforms must balance security, usability, functional benefits, and user enjoyment to effectively attract and retain users. These variables integrate both hedonic and utilitarian aspects of user evaluations, offering a balanced perspective. While not exhaustive, they represent the most salient factors influencing user ambivalence and retention intention in this context.

Although prior research has explored the adverse association between ambivalence and behavioral intentions—such as adoption intentions in online dating services or avoidance behaviors on social networking sites—its relationship with user retention in online hotel booking platforms remains underexplored. This gap positions H1 as a novel contribution within the context of online hotel booking. The remaining hypotheses (H2–H6) extend and enrich the broader IS literature by offering novel insights into how various cognitive and emotional antecedents are associated with user ambivalence and, in turn, relate to retention outcomes.

3.2. Consequences of ambivalence

Evaluations plays a critical role in guiding individuals' behaviors (Van Harreveld et al., 2015). According to Cognitive Dissonance Theory (Festinger, 1957), conflicting or inconsistent evaluations often give rise to ambivalence, a state characterized by the simultaneous presence of positive and negative associations. Ambivalence generates psychological discomfort and unpleasantness (Van Harreveld et al., 2015), which is adversely related to behavioral intentions. This discomfort, or dissonance, motivates individuals to resolve the inconsistency by altering their cognitions, behaviors, or the perceived importance of the conflict (Festinger, 1957).

In the context of IS use, ambivalence often arises when users experience trade-offs between a system's features (Ashforth et al., 2014; Ng et al., 2021). For example, users may perceive a system as useful and enjoyable while simultaneously finding it as effort-intensive or posing security risks. The discomfort caused by these conflicting evaluations prompts coping strategies (Qahri-Saremi and Turel, 2020), such as

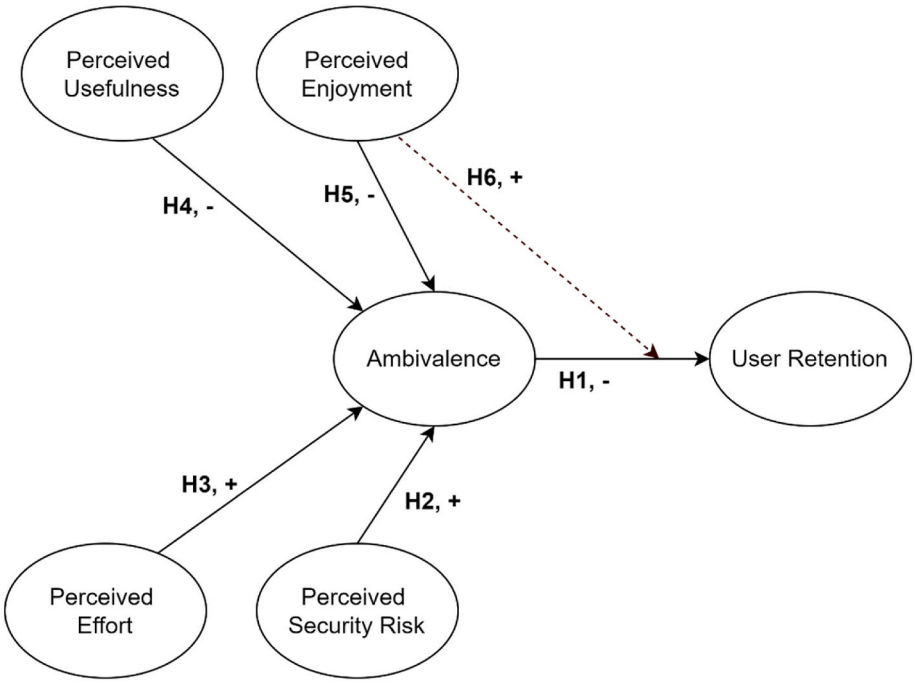


Fig. 1. Ambivalence and user retention model in online hotel booking.

reducing usage or discontinuing the system.

In online hotel booking, users are particularly prone to ambivalence when they encounter both benefits (e.g., usefulness and enjoyment) and drawbacks (e.g., security risks and high effort). The cognitive dissonance that arises during decision-making heightens discomfort and cognitive load, pushing users to seek ways to resolve these conflicts. However, when platforms present significant trade-offs, they may fail to adequately alleviate dissonance, resulting in a negative user experience. Over time, the discomfort caused by ambivalence is likely to diminish user satisfaction, lower engagement, and ultimately harm user retention. Therefore, the following hypothesis is proposed.

H1. Ambivalence is negatively associated with user retention in online hotel booking.

3.3. Perceived security risk and ambivalence

Perceived security risk originates from risk perception theory (Slovic, 1987), which suggests that individuals assess potential threats or dangers through their subjective interpretation of available information. In online environments, security risk perceptions are heightened due to the intangibility of services, the sensitivity of shared information, and the potential for harm, such as identity theft or financial loss (Featherman and Pavlou, 2003). These perceptions have long been recognized as barriers to technology adoption (Franque et al., 2021) and online booking behaviors (San-Martín et al., 2020).

In the context of online hotel booking, higher perceived security risk is likely associated with greater user ambivalence, as it triggers negative evaluations—such as concerns about fraud or data breaches—that conflict with positive evaluations, such as the platform's convenience and usefulness. This evaluative conflict fosters ambivalence, as users struggle to reconcile the potential for harm with the perceived benefits of using the platform. The resulting tension underscores the user's difficulty in resolving these conflicting beliefs, ultimately intensifying their overall ambivalence. Therefore, the following hypothesis is proposed.

H2. Perceived security is positively associated with user ambivalence in online hotel booking.

3.4. Perceived effort and ambivalence

Perceived effort refers to the extent to which users anticipate that using a system will be challenging or require significant mental or physical resources (Davis, 1989). According to Resource Allocation Theory (Kahneman, 1973), individuals have a limited pool of cognitive and physical resources that must be allocated across various activities. When using a system is perceived as effortful, it demands a larger share of these limited resources, increasing cognitive strain and reducing their availability for other tasks (Davis, 1989).

In the context of online hotel booking, high perceived effort is a significant negative factor that harms user experience. Tasks requiring substantial effort in the booking process heighten users' cognitive load, resulting in frustration and discomfort. These negative evaluations conflict with the platform's perceived benefits, such as usefulness or convenience, leading to a state of ambivalence—a coexistence of positive and negative assessments. Users experience tension as they attempt to reconcile the cognitive burden associated with using the platform against its perceived advantages. Therefore, the following hypothesis is proposed.

H3. Perceived effort is positively associated with user ambivalence in online hotel booking.

3.5. Perceived usefulness and ambivalence

Utilitarian motivation theory (Davis, 1989; Davis et al., 1989) posits that individuals are primarily motivated by the functional and practical benefits a system provides, emphasizing its goal-oriented and instrumental value. Unlike hedonic motivation, which focuses on pleasure and enjoyment, utilitarian motivation drives individuals to use systems to achieve tangible, efficient outcomes that satisfy their needs with minimal effort and maximum benefit (Davis, 1989; Davis et al., 1989).

Perceived usefulness, a central construct in this theory, refers to the extent to which users believe a system enhances their task performance or helps them achieve specific goals (Davis, 1989). In technology adoption, perceived usefulness reflects a system's instrumental value, enabling users to meet their objectives more effectively and efficiently (Davis, 1989; Davis et al., 1989). Users who perceive a system as highly useful are more likely to appreciate its functional benefits, resulting in favorable evaluations and increased satisfaction (Franque et al., 2021).

For online hotel booking platforms, high perceived usefulness indicates that the platform effectively assists users in achieving booking-related tasks, such as finding the best deals, securing reservations quickly, or accessing relevant information with ease. This aligns with utilitarian motivations, where users are goal-directed and focused on achieving specific outcomes efficiently (Davis et al., 1992). When users perceive the platform as highly useful, they tend to focus on its functional value, which helps reduce cognitive dissonance by diminishing conflicting evaluations. The platform's clear practical benefits offset negative assessments, thereby reducing ambivalence. Therefore, the following hypothesis is proposed.

H4. Perceived usefulness is negatively associated with user ambivalence in online hotel booking.

3.6. Perceived enjoyment and ambivalence

Perceived enjoyment refers to the extent to which a system's use is intrinsically pleasurable, independent of any performance-related outcomes (Davis et al., 1992; Venkatesh, 2000). According to Intrinsic Motivation Theory (Deci and Ryan, 1985), individuals engage in activities for the inherent pleasure they derive from the activity itself. Enjoyable experiences fulfill intrinsic psychological needs, such as pleasure, autonomy, and competence, which lower the likelihood of experiencing mixed or contradictory feelings (Deci and Ryan, 1985). Building on user experience research, enjoyable interactions generate positive emotions that tend to overshadow negative feelings, thereby mitigating the discomfort associated with cognitive or emotional conflict.

In the context of online hotel booking, perceived enjoyment enables users to focus on the platform's positive aspects—such as ease of use, efficiency, and convenience—while placing less emphasis on potential drawbacks, like security risks. By fostering positive affect, greater enjoyment helps reduce cognitive and emotional conflict, thereby being associated with lower ambivalence. The pleasurable experience provided by the platform neutralizes or overrides conflicting evaluations, leading to more consistent and favorable perceptions. Consequently, users who derive intrinsic enjoyment from the platform are less likely to experience tension between positive and negative assessments, ultimately experiencing a lower degree of ambivalence. Therefore, the following hypothesis is proposed.

H5. Perceived enjoyment is negatively associated with user ambivalence in online hotel booking.

3.7. The moderating role of perceived enjoyment

In the user experience (UX) literature, perceived enjoyment is closely associated with positive affect and emotional engagement, both of which play a critical role in shaping users' overall evaluations of a system (Hassenzahl, 2010). Enjoyable experiences fulfill intrinsic psychological needs—such as autonomy, competence, and relatedness—thereby fostering heightened satisfaction and emotional connection (Deci and Ryan, 1985). Research has shown that perceived enjoyment

enhances user engagement and reduces attention to negative factors, such as frustration or perceived complexity (Turel and Serenko, 2012). High levels of perceived enjoyment can lead to a flow state (Agarwal and Karahanna, 2000), in which users become fully absorbed in the activity, experiencing deep engagement and emotional gratification. In this state, cognitive and emotional focus shifts toward the system's positive aspects, diminishing the salience of negative evaluations.

In the context of online hotel booking, perceived enjoyment enables users to focus on delightful platform aspects, such as intuitive design, seamless navigation, and visually appealing interfaces, while de-emphasizing drawbacks (Bilgihan et al., 2015; Hassenzahl, 2008). Such positive emotional experiences help mitigate cognitive conflict and override the discomfort caused by mixed evaluations. Thus, perceived enjoyment serves as a buffer against ambivalence by amplifying emotional satisfaction and minimizing conflicting feelings. Users who find the booking platform enjoyable are less likely to dwell on negative aspects, as the pleasurable experience dominates their overall evaluation and encourages consistent engagement. Therefore, the following hypothesis is proposed.

H6. Perceived enjoyment moderates the negative relationship between ambivalence and user retention in online hotel booking, such that higher levels of enjoyment are associated with a weaker negative relationship between ambivalence and user retention.

4. Methodology

4.1. Survey administration

An online survey was developed using Qualtrics to collect data on user experiences and retention with online hotel booking platforms. The survey design employed established measurement scales from prior literature to ensure validity and reliability. All research variables were operationalized using multiple items adapted from existing studies, as summarized in Table 2. Constructs, including perceived security risk, effort, usefulness, enjoyment, and user retention were measured on a 7-point Likert scale, ranging from 1 ("strongly disagree") to 7 ("strongly agree").

Ambivalence was assessed using differential semantic scales to capture its positive and negative components, with each scale ranging from 1 ("not at all") to 5 ("extremely positive" or "extremely negative," respectively). The magnitude of ambivalence was calculated using the Griffin formula:

$$\text{Ambivalence} = [(\text{Positive} + \text{Negative})/2] - |\text{Positive} - \text{Negative}|$$

The formula accounts for both the intensity of positive and negative feelings and the degree of conflict between them (Thompson et al., 1995). Its application aligns with recent IS ambivalence research, which has successfully used this approach to measure ambivalence in digital contexts (Dang and Liu, 2021; Lakhiwal et al., 2023; Ng et al., 2021).

To ensure clarity and consistency, the questionnaire was reviewed by four researcher experts, and their feedback was incorporated to refine item wording and enhance the overall survey structure. The finalized survey was deployed via Amazon Mechanical Turk (MTurk) to recruit participants. Respondents were required to specify an online hotel booking platform they had used and answer questions based on their actual experiences. Screening questions ensured that participants were at least 18 years old and had recent experience with online hotel booking platforms.

4.2. Sample and data collection

A total of 500 participants were initially recruited through MTurk to participate in the online survey. To ensure data integrity and reliability, rigorous procedures were implemented following best practices for PLS-SEM analysis (Hair et al., 2022). Three responses with incomplete data

Table 2
Conceptualization of constructs.

Construct and items	References
Perceived Security Risk 1. Using the travel website to book hotels exposes me to fraud or monetary loss. 2. The travel website makes me feel unsafe when using it to book hotels. 3. Using the travel website to book hotels leads to a loss of privacy because my personal information is used without consent.	Lee (2009)
Perceived Effort 1. The use of the travel website to book hotels takes too much time. 2. The use of the travel website to book hotels requires too much effort. 3. The use of the travel website to book hotels is too complex.	(Davis, 1989)
Perceived Usefulness 1. Using the travel website improves my performance in booking hotels. 2. Using the travel website increases my productivity in booking hotels. 3. Using the travel website enhances my effectiveness in booking hotels. 4. I find that the travel website is useful for booking hotels.	(Davis, 1989; Davis et al., 1989)
Perceived Enjoyment 1. I find that using the travel website to book hotels is enjoyable. 2. The process of using the travel website to book hotels is pleasant. 3. I have fun when using the travel website to book hotels. 4. I find that using the travel website to book hotels is interesting.	(Davis et al., 1992; Venkatesh, 2000)
Ambivalence *Positive component* 1. Regardless of how dissatisfied you feel, to what extent do you feel satisfied with booking hotels on the travel website? 2. Regardless of any negative feelings, how positive do you feel about booking hotels on the travel website? 3. Regardless of any perceived disadvantageous, to what extent do you find the travel website advantageous for booking hotels? *Negative component* 1. Regardless of how satisfied you feel, to what extent do you feel dissatisfied with booking hotels on the travel website? 2. Regardless of any positive feelings, how negative do you feel about booking hotels on the travel website? 3. Regardless of any perceived advantages, to what extent do you find the travel website disadvantageous for booking hotels?	(Thompson et al., 1995; Yu and Davis, 2024)
User Retention Intention 1. I intend to continue using the travel website to book hotels in the future. 2. I plan to keep using the travel website to book hotels in my daily life. 3. I will continue using the travel website to book hotels as frequently as I currently do.	(Bhattacherjee, 2001)

on key constructs were excluded, and 110 responses exhibiting suspicious patterns, such as straight-lining or survey completion in under five minutes, were removed based on established recommendations (Lee and Park, 2024). These measures enhanced the quality and reliability of the dataset.

The final sample consisted of 387 participants, representing 77.4% of the initial responses. This sample size ensures adequate statistical power for the analysis and meets criteria recommended by prior research (Hair et al., 2022; Roy and Shaikh, 2024). According to Hair et al. (2022), the minimum sample size for PLS-SEM should be at least 10 times the maximum number of arrowheads pointing to any latent variable in the path model. In our model, the construct with the highest number of arrowheads, user retention, has five, requiring a minimum of 50 participants. Furthermore, a G*Power analysis confirmed that a sample size of 64 participants would achieve a statistical power of 0.8 for detecting a medium effect size ($f^2 = 0.5$) (Roy and Shaikh, 2024). With 387 participants, our sample size significantly exceeds these thresholds, ensuring robust statistical power, reliable model estimation, and a strong foundation for valid conclusions.

As shown in Table 3, the sample represents a diverse group of online hotel booking platform users. The gender distribution was relatively balanced, with 57.4% identifying as male and 42.6% as female. The majority of participants were younger, with 38.8% aged 18–29 and 37.7% aged 30–39, while 13.4% were aged 40–49, and 10.1% were above 50. Additional demographic details, including participants' education levels, duration of platform use, and usage frequency, are provided in Table 3.

Participants reported using a variety of online hotel booking platforms, including Expedia (22%), Hotels.com (12.1%), TripAdvisor (9.3%), Bookings.com (8%), Travelocity (6.5%), Kayak (4.9%), Priceline (4.7%), and other platforms (32.5%).

Table 3
Sample characteristics (N = 387).

Category		Frequency	Percentage
Gender	Male	223	57.4%
	Female	164	42.6%
Age	18–29	150	38.8%
	30–39	146	37.7%
	40–49	52	13.4%
	Above 50	39	10.1%
Education levels	High school or less	15	3.9%
	Some college	70	18.1%
	Associate degree	38	9.8%
	Bachelor's degree	174	45.0%
	Master's degree	68	17.6%
	Doctorate	9	2.3%
	Professional degree	13	3.4%
How long have you been using the hotel booking platform?	Less than six months	44	11.4%
	Six months to one year	105	27.2%
	One year to three years	152	39.4%
	Three to five years	54	14.0%
	Over five years	31	8.0%
How frequently have you been using the hotel booking platform?	Several times a week	4	1.0%
	Once a week	22	5.7%
	Once a month	84	21.7%
	Every few months	226	58.4%
	Once a year	51	13.2%

5. Data analysis and results

The proposed model and hypotheses were tested using the variance-based partial least squares structural equation model (PLS-SEM) approach with SmartPLS 4.0 software (Ringle et al., 2024). PLS-SEM was chosen over covariance-based SEM (CB-SEM) due to its robustness and suitability for exploratory research, particularly when theoretical development is ongoing or insufficiently established (Hair et al., 2011, 2022). This study focuses on ambivalence in online hotel booking, a relatively nascent area, and prioritizes predictive accuracy over theory confirmation. PLS-SEM is well-suited for examining complex models that include direct, indirect, and moderating relationships, making it ideal for this research. Additionally, the approach accommodates data that may deviate from normality and is effective for small to moderate sample sizes. By focusing on maximizing explained variance in endogenous constructs, PLS-SEM facilitates theoretical advancements in emerging research domains (Hair et al., 2011). By emphasizing prediction and offering flexibility in addressing normality constraints, PLS-SEM enabled the testing of a comprehensive model of ambivalence, yielding valuable insights into the interplay between cognitive and emotional conflicts and their impact on user retention in our study. This methodological choice ensured the reliability and robustness of the findings while advancing understanding in this underexplored domain.

5.1. Common method bias assessment

Before proceeding with the analysis, we assessed common method bias to ensure it would not significantly affect the findings. We first conducted Harman's single-factor test (Podsakoff et al., 2003) on the six constructs in our model: perceived usefulness, enjoyment, effort, security risk, ambivalence, and user retention. The largest variance explained by a single factor was 39%, well below the 50% threshold, suggesting minimal concern for common method bias. Additionally, we employed a single-method factor approach within PLS (Liang et al., 2007) for a more rigorous assessment. This method compares each indicator's variance explained by its primary construct with the variance explained by a common method factor. Following the approach outlined by Liang et al. (2007), we incorporated a common method factor to the PLS model, using all primary construct indicators as its indicators. The results showed that all substantive variances (those explained by primary constructs) exceeded method variances (those explained by the common method factor), with substantive-to-method variance ratio of 162:1. This ratio meets the recommended criteria in prior literature (Hew et al., 2018; Liang et al., 2007). Based on these assessments, common method bias is unlikely to pose a significant concern in our study.

5.2. Measurement model evaluation

To evaluate the measurement model, we followed established recommendations (Fornell and Larcker, 1981; Hair et al., 2022; Henseler et al., 2015) by assessing reliability and convergent validity using Cronbach's alpha, composite reliability, and Average Variance Extracted (AVE).

As shown in Table 4, all factor loadings ranged from 0.80 to 0.97, exceeding the recommended threshold of 0.7 (Hair et al., 2022), indicating sufficient indicator reliability. Table 5 demonstrates that Cronbach's alpha values ranged from 0.88 to 0.95, and composite reliability values ranged from 0.92 to 0.97, both well above the threshold of 0.7 (Fornell and Larcker, 1981; Hair et al., 2022). These results confirm sufficient internal consistency reliability. Additionally, as shown in Table 5, AVE values ranged from 0.74 to 0.92, suggesting that each construct accounted for more than half of the variance in its items. This meets the recommended criterion of 0.5 (Henseler et al., 2015; Patrizi et al., 2024), thereby reflecting adequate convergent validity.

To confirm discriminant validity, we compared the correlations

Table 4
Item loadings and cross-loadings.

	PU	PE	PEF	PSR	AMB	URI
PU1	0.87	0.46	−0.18	−0.24	−0.36	0.51
PU2	0.87	0.47	−0.20	−0.23	−0.33	0.52
PU3	0.89	0.47	−0.20	−0.22	−0.39	0.51
PU4	0.80	0.39	−0.29	−0.32	−0.41	0.51
PE1	0.48	0.92	−0.05	−0.07	−0.24	0.44
PE2	0.51	0.89	−0.13	−0.13	−0.28	0.46
PE3	0.38	0.86	0.07	0.07	−0.18	0.37
PE4	0.46	0.89	0.00	0.00	−0.20	0.43
PEF1	−0.25	−0.05	0.95	0.63	0.43	−0.25
PEF2	−0.24	−0.03	0.97	0.66	0.47	−0.24
PEF3	−0.25	−0.03	0.95	0.66	0.45	−0.25
PSR1	−0.27	−0.05	0.61	0.93	0.47	−0.23
PSR2	−0.26	−0.02	0.66	0.95	0.49	−0.24
PSR3	−0.30	−0.06	0.64	0.94	0.50	−0.25
AMB1	−0.40	−0.23	0.40	0.44	0.90	−0.43
AMB2	−0.39	−0.26	0.44	0.46	0.91	−0.44
AMB3	−0.38	−0.20	0.44	0.50	0.89	−0.44
URI1	0.57	0.41	−0.29	−0.29	−0.47	0.93
URI2	0.53	0.49	−0.17	−0.17	−0.41	0.89
URI3	0.55	0.42	−0.25	−0.25	−0.46	0.93

Note: PU = Perceived Usefulness, PE = Perceived Enjoyment, PEF = Perceived Effort, PSR= Perceived Security Risk, AMB = Ambivalence, and URI = User Retention Intention.

Table 5
Descriptive statistics, internal reliability, and convergent validity.

Construct name	Mean	SD	Cronbach's alpha (>0.70)	Composite reliability (>0.70)	AVE values (>0.50)
PU	5.66	0.86	0.88	0.92	0.74
PE	5.22	1.08	0.91	0.94	0.79
PEF	2.89	1.57	0.95	0.97	0.92
PSR	2.69	1.50	0.93	0.96	0.88
AMB	0.62	1.22	0.88	0.93	0.81
URI	5.71	1.08	0.90	0.94	0.84

Note: 1. PU = Perceived Usefulness, PE = Perceived Enjoyment, PEF = Perceived Effort, PSR= Perceived Security Risk, AMB = Ambivalence, and URI = User Retention Intention; 2. Ambivalence was calculated using Griffin's formula: AMB = (PA + NA)/2 -| PA-NA|. PA and NA each range from 1 (not at all) to 5 (extremely positive or negative), and the AMB score ranges from −1 to 5. User retention intention, perceived security risk, effort, usefulness, and enjoyment were all measured on 7-point Likert scales, ranging from 1 (strongly disagree) to 7 (strongly agree).

among all constructs with the square roots of their respective AVE values (Fornell and Larcker, 1981). As shown in the triangular matrix in Table 6, the diagonal values (i.e., square roots values of AVE) are greater than the inter-construct correlations, confirming discriminant validity. Additionally, the Heterotrait-Monotrait (HTMT) ratio was used as a secondary test (Henseler et al., 2015). As shown in Table 7, all HTMT values were below the recommended threshold of 0.9, further affirming discriminant validity.

Table 6
Inter-construct correlation matrix.

	PU	PE	PEF	PSR	AMB	URI
PU	**0.86**					
PE	0.52	**0.89**				
PEF	−0.26	−0.04	**0.96**			
PSR	−0.30	−0.05	0.68	**0.94**		
AMB	−0.44	−0.26	0.47	0.52	**0.90**	
URI	0.60	0.48	−0.26	−0.26	−0.48	**0.92**

Note: 1. The diagonal elements in bold represent the squared root of the AVE, while the off-diagonal elements indicate the correlations between constructs; 2. PU = Perceived Usefulness, PE = Perceived Enjoyment, PEF = Perceived Effort, PSR= Perceived Security Risk, AMB = Ambivalence, and URI = User Retention Intention.

Table 7
Discriminant validity — HTMT Scores.

	PU	PE	PEF	PSR	AMB	URI
PU						
PE	0.58					
PEF	0.28	0.08				
PSR	0.33	0.09	0.72			
AMB	0.49	0.28	0.52	0.57		
URI	0.67	0.53	0.28	0.28	0.54	

Note: 1. Obtained HTMT ratio values were below the recommended threshold of 0.9; 2. PU = Perceived Usefulness, PE = Perceived Enjoyment, PEF = Perceived Effort, PSR= Perceived Security Risk, AMB = Ambivalence, and URI = User Retention Intention.

5.3. Model fit

Before evaluating the model, we tested for multicollinearity by examining the Variance Inflation Factor (VIF) values for the constructs. The obtained VIF values ranged between 1.1 and 1.9, well below the recommended threshold of 3.0 (Hair et al., 2022), indicating that multicollinearity was not a concern in our study.

We then assessed the overall model fit. The Standardized Root Mean Square Residual (SRMR) value was 0.065 in our study, which is below the recommended threshold of 0.08 (Henseler et al., 2015), indicating a good model fit. To further evaluate the model's predictive power, we examined the R^2 and Q^2 values for key constructs. The adjusted R^2 values were 0.39 for ambivalence and 0.40 for user retention, both exceeding the commonly accepted threshold of 0.2 (Hair et al., 2011), suggesting acceptable explanatory power. Additionally, we assessed the Q^2 values to evaluate the model's predictive relevance. The obtained Q^2 values were 0.35 for ambivalence and 0.41 for retention, both well above the threshold of 0 (Lu and Shi, 2024). These findings suggest that the model exhibits strong explanatory and predictive capabilities.

5.4. Hypothesis testing

To evaluate the significance of the path coefficients in the structural model, we used a bootstrapping procedure with 5000 subsamples (Hair et al., 2022). Path coefficients (β), t-values, 95% confidence intervals, and significance levels were calculated and summarized in Table 8. A visual representation of model testing results is provided in Fig. 2.

As shown in Table 8, the results indicate that ambivalence is negatively associated with user retention (β = −0.410, t = 10.408, p <

Table 8
Hypothesis testing results.

No.	Hypotheses	β	t	p	95% CIs biased corrected	Results
H1	AMB → URI	−0.410	10.408	<0.001	[-0.483; −0.330]	Supported
H2	PSR → AMB	0.307	4.935	<0.001	[0.186; 0.428]	Supported
H3	PEF→ AMB	0.201	3.358	0.001	[0.085; 0.319]	Supported
H4	PU → AMB	−0.236	4.573	<0.001	[-0.334; −0.130]	Supported
H5	PE → AMB	−0.111	2.584	0.010	[-0.196; −0.027]	Supported
H6	PE × AMB → URI	0.162	3.296	0.001	[0.067; 0.259]	Supported

Note: 1. H1 is novel in online booking context; H2 - H6 are novel contributions to the broader IS literature. 2. PU = Perceived Usefulness, PE = Perceived Enjoyment, PEF = Perceived Effort, PSR= Perceived Security Risk, AMB = Ambivalence, and URI = User Retention Intention.

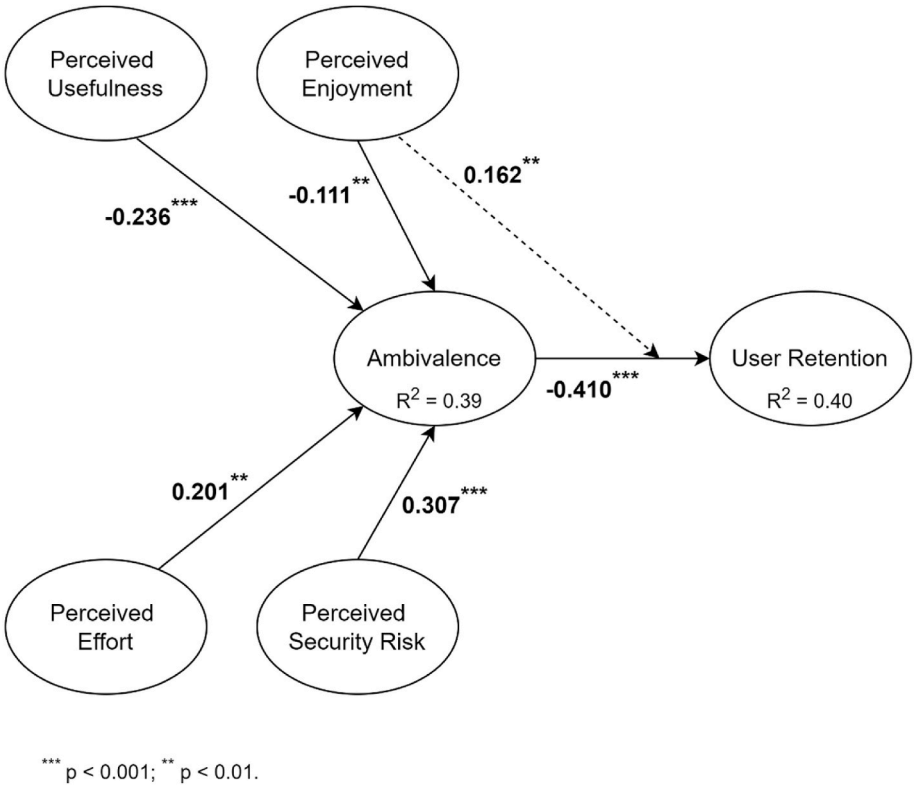


Fig. 2. Ambivalence and user retention model results.

0.001), supporting H1. Perceived security risk is positively associated with ambivalence (β = 0.307, t = 4.935, p < 0.001), supporting H2. Similarly, perceived effort is positively associated with ambivalence (β = 0.201, t = 3.358, p < 0.01), thereby supporting H3. Table 8 also shows that both perceived usefulness and enjoyment are negatively associated with ambivalence (β = −0.236, t = 4.573, p < 0.001, and β = −0.111, t = 2.584, p < 0.01, respectively), supporting H4 and H5. The moderating effect of perceived enjoyment, as described in H6, is discussed in the next section.

5.5. Moderation analysis

To test the moderating role of perceived enjoyment in the relationship between ambivalence and user retention in online booking, we employed the product indicator approach in SmartPLS (Ringle et al., 2024) to create an interaction term between the moderator and the independent variable (i.e., perceived enjoyment × ambivalence). As shown in Table 8, the obtained p-value of 0.001 from bootstrapping confirmed the significance of the moderating effect. Additionally, the confidence interval did not include zero, further validating the interaction effect and the moderating role of perceived enjoyment.

Following established guidelines (Hair et al., 2022; Patrizi et al., 2024), we conducted a simple slope analysis in SmartPLS to visualize the moderation effect. As shown in Fig. 3, the middle line (dashed) represents the relationship between ambivalence (x-axis) and user retention (y-axis) at an average level of perceived enjoyment (mean level). The lower line (dotted) represents this relationship at a lower level of perceived enjoyment (−1 SD), while the upper line (solid) represents it at a higher level of perceived enjoyment (+1 SD). The less steep slope observed at higher levels of perceived enjoyment indicates that the negative association between ambivalence on user retention is mitigated when perceived enjoyment is higher.

In addition to the above simple slope analysis, we assessed the f^2 effect size of the interaction effect to provide a more comprehensive evaluation of the moderation results (Hair et al., 2022). The obtained f^2 effect size was 0.037, indicating a large moderating effect based on the recommend criteria of 0.005 (small), 0.01 (medium), and 0.025 (large) (Kenny, 2018).

Overall, these findings confirm that perceived enjoyment significantly moderates the relationship between ambivalence and user retention, thus supporting H6. Specifically, when users perceive their online booking platform as highly enjoyable, the negative association between ambivalence and user retention is weaker, demonstrating that enjoyment mitigates the detrimental impact of ambivalence on retention.

5.6. Mediation analysis

We conducted a mediation analysis to examine the mediating role of ambivalence in the relationship between its antecedents and user intention, following established procedures (Rodríguez-Priego et al., 2023; Zhao et al., 2010). A bootstrap method with 5000 resamples was employed to estimate 95% confidence intervals for the indirect effect, with a summary of the results presented in Table 9.

The results indicate that for perceived usefulness and perceived enjoyment, the indirect effects through ambivalence are significant (β = 0.097, p < 0.001 and β = 0.046, p < 0.01, respectively), while their direct effects on retention remain significant (β = 0.340, p < 0.001 and β = 0.233, p < 0.001, respectively). These findings suggest that ambivalence plays a partial mediating role in these relationships. In contrast, the indirect effects of perceived effort and perceived security risk on user retention are significant (β = −0.082, p < 0.01 and β = −0.126, p < 0.001, respectively), while their direct effects are not significant (β = −0.061, p > 0.05 and β = 0.021, p > 0.05, respectively). This indicates that ambivalence fully mediates the relationships between these two antecedents and retention.

These findings highlight the critical role of ambivalence in shaping the relationship between platform features and user retention. Specifically, while ambivalence fully mediates the influence of perceived security risk and effort, it partially explains the effects of perceived

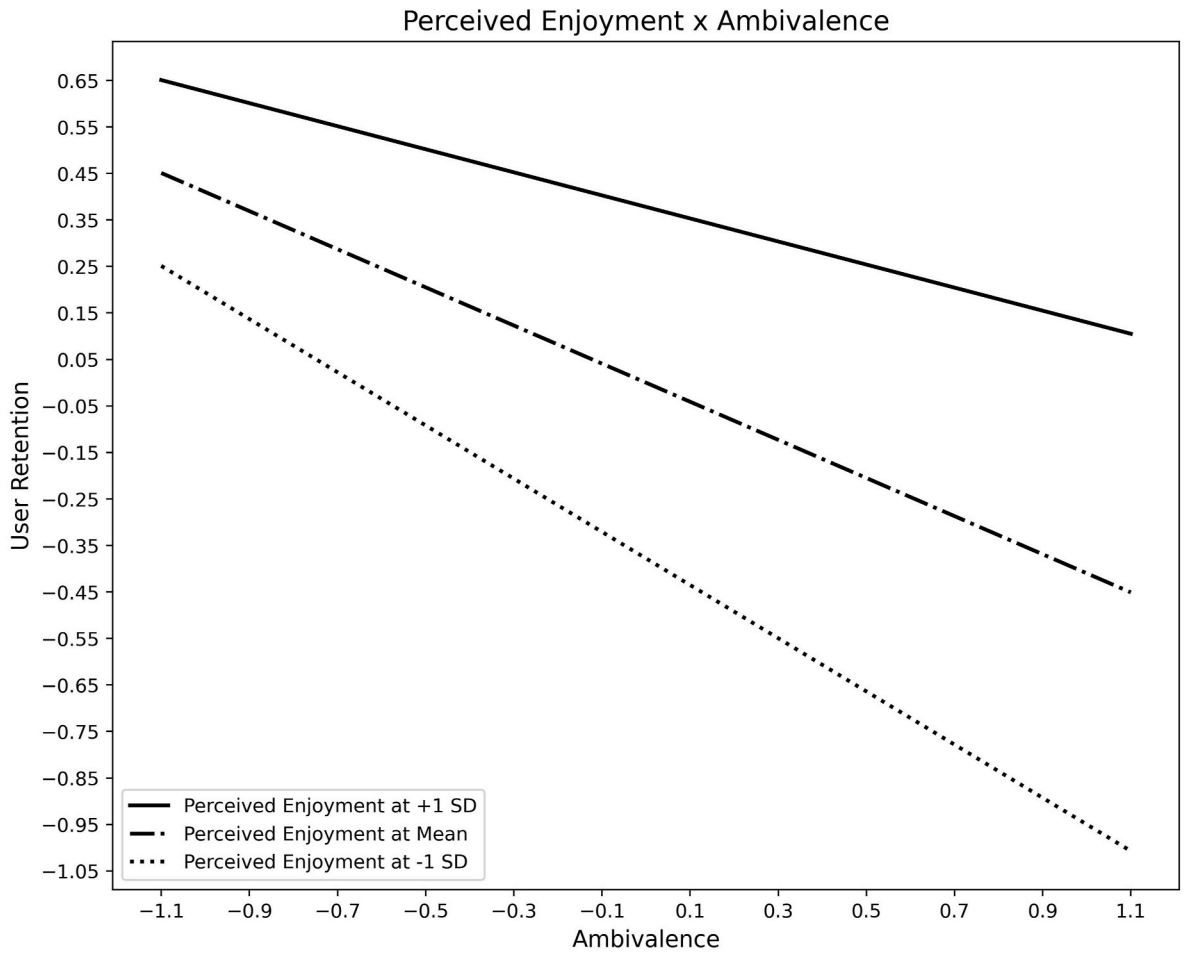


Fig. 3. Moderation analysis: Simple slope plot.

Table 9
Mediation analysis results.

Relationship	Indirect effect	95% CIs	t	p	Mediation Type
PU → AMB → URI	0.097	[0.050, 0.150]	3.804	<0.001	Partial
PE → AMB → URI	0.046	[0.012, 0.080]	2.607	0.009	Partial
PEF → AMB → URI	−0.082	[-0.137, −0.035]	3.212	0.001	Full
PSR → AMB → URI	−0.126	[-0.185, −0.075]	4.461	<0.001	Full

Note: PU = Perceived Usefulness, PE = Perceived Enjoyment, PEF = Perceived Effort, PSR= Perceived Security Risk, AMB = Ambivalence, and URI = User Retention Intention.

usefulness and enjoyment, highlighting its multifaceted impact on user retention.

6. Discussion

6.1. Key findings

This study provides significant insights into the relationships among antecedent beliefs, ambivalence, and user retention, advancing the understanding of post-adoption behaviors in online hotel booking platforms. By extending prior research on ambivalence in other technology contexts (Chen et al., 2023; Lakhiwal et al., 2023; Qahri-Saremi and Turel, 2020; Turel and Qahri-Saremi, 2023), our findings highlight ambivalence as a critical factor influencing user engagement in high-stakes, decision-intensive settings like online hotel booking.

Our results reveal that ambivalence is negatively associated with user retention. Drawing on Ambivalence theory (Thompson et al., 1995) and Cognitive Dissonance Theory (Festinger, 1957), we demonstrate that ambivalence induces psychological discomfort, arising from conflicting positive (e.g., perceived usefulness, enjoyment) and negative (e.g., perceived security risks, high effort) evaluations. This cognitive tension creates internal conflict, motivating users to disengage from the platform as a way to alleviate the discomfort of mixed feelings. Consequently, users experiencing higher levels of ambivalence are less likely to remain committed to the platform, emphasizing ambivalence's critical role in shaping retention outcomes.

The relationship between antecedent beliefs and ambivalence reveals key psychological mechanisms. Concerns about security risks generate cognitive and emotional conflict. While users value the online booking platform's convenience, security concerns undermine their evaluations. This aligns with Risk Perception Theory (Slovic, 1987), which explains how users weigh potential harm against perceived benefits, leading to evaluative tension. High perceived effort increases ambivalence by imposing cognitive load on users. Resource Allocation Theory (Kahneman, 1973) suggests that cognitive resources are limited; when booking processes require significant effort, frustration arises, exacerbating conflicting evaluations. Perceived usefulness reduces ambivalence by aligning the platform with users' task-oriented goals. According to Utilitarian Motivation Theory (Davis, 1989), instrumental value minimizes cognitive dissonance and reduces the weight of negative evaluations. Perceived enjoyment alleviates ambivalence by fostering positive affect that overshadows negative evaluations. Intrinsic

Motivation Theory (Deci and Ryan, 1985) suggests that enjoyable experiences satisfy intrinsic psychological needs, reducing cognitive tension and promoting consistent evaluations.

A particularly novel finding is the moderating effect of perceived enjoyment. While prior research primarily positions enjoyment as a direct driver of behavioral intentions (Davis et al., 1992; Venkatesh, 2000), our results show that perceived enjoyment also buffers the negative association between ambivalence and retention. When perceived enjoyment is high, users are more likely to tolerate conflicting evaluations. Enjoyable experiences dominate users' focus, reducing the salience of negative aspects and mitigating ambivalence's detrimental influence on retention.

Another novel finding is that ambivalence mediates the relationship between antecedent beliefs and retention, highlighting its central role as a psychological mechanism. For perceived security risk and effort, ambivalence fully mediates the relationship with retention. These antecedents influence retention primarily through the cognitive and emotional conflict they generate. For perceived usefulness and enjoyment, ambivalence partially mediates the relationship with retention. These factors are not only associated with ambivalence but also directly linked to user retention. This dual mediation mechanism underscores the complex pathways through which antecedent beliefs shape retention outcomes.

6.2. Theoretical implications

This study makes several significant contributions to the IS literature by advancing the understanding of ambivalence in digital user behavior, particularly its role in post-adoption contexts and platform retention. By positioning ambivalence as a central construct, this research offers a nuanced perspective that captures the coexistence of positive and negative evaluations, enriching broader theories of user engagement and loyalty.

Our findings highlight ambivalence as an intermediary between antecedent beliefs and retention. Traditional engagement and loyalty models (e.g., Bhattacherjee, 2001) focus predominantly on positive evaluations and their associations with retention. This study, however, demonstrates that ambivalence—a state of cognitive and emotional conflict—relates negatively to retention. By addressing both positive and negative evaluations, this study adds balance to theories of user behavior (Park et al., 2021), emphasizing the significance of mixed evaluations in digital environments (Khalek et al., 2025).

This research aligns with the bivariate unipolar perspective proposed by Yu and Davis (2024), which suggests that positive and negative evaluations such as satisfaction and dissatisfaction can coexist and collectively relate to behavioral intention. This challenges traditional bipolar satisfaction models (e.g., Bhattacherjee, 2001), which position satisfaction and dissatisfaction as two opposing ends of a single bipolar continuum. By adopting this nuanced view, the study provides a deeper understanding of user behavior in complex decision-making environments.

A significant contribution of this study is the moderating role of perceived enjoyment. Previous research primarily positions perceived enjoyment as directly linked to behavioral intentions (Davis et al., 1992; Venkatesh, 2000). This study, however, indicates that perceived enjoyment moderates the association between ambivalence and retention, such that when perceived enjoyment is high, the negative association between ambivalence and retention weakens. By fostering positive affect, perceived enjoyment may reduce the prominence of negative evaluations, thereby enriching user experience literature (Hassenzahl and Tractinsky, 2006) and advancing the understanding of emotional engagement in digital environments.

Beyond perceived enjoyment, this study suggests that variables such as familiarity and trust may moderate the associations between ambivalence and retention. Familiarity with a platform could reduce uncertainty and strengthen positive evaluations, while trust may lessen concerns tied to security risks. Future research could further investigate these variables to better understand how contextual and individual factors interact with ambivalence and retention outcomes.

This research builds on TAM (Davis, 1989) and UTAUT (Venkatesh et al., 2003) by demonstrating that perceived effort relates not only to adoption challenges but also to ambivalence. Additionally, the findings contribute to Cognitive Dissonance Theory (Festinger, 1957) by illustrating how users with conflicting evaluations of effort and usefulness may balance these evaluations when considering whether to continue using a platform. For example, users perceiving high effort may still engage with the platform if it is perceived as highly useful, reflecting the complexity of user decision-making in the presence of ambivalence.

This study highlights perceived security risk as an important factor associated with ambivalence. Security concerns appear to relate to cognitive and emotional conflict, which is reflected in retention outcomes. By exploring the link between security risk and ambivalence, this research extends Risk Perception Theory (Slovic, 1987) and underscores the value of trust-building measures in platform design to address ambivalence and improve user retention.

By focusing on a high-stakes, decision-intensive context, this study extends ambivalence theory (Thompson et al., 1995) to online hotel booking. While previous studies primarily address ambivalence in social networking or hedonic platforms (Chen et al., 2023; Qahri-Saremi and Turel, 2020; Turel and Qahri-Saremi, 2023), this study explores ambivalence stemming from cognitive and emotional conflicts related to security, effort, and usability. This contextual lens enriches ambivalence theory by emphasizing its relevance for understanding user behavior in complex digital environments.

6.3. Practical implications

This study provides actionable insights for platform designers, product managers, and marketers in the online hotel booking industry to address user ambivalence effectively. By addressing factors contributing to ambivalence—such as perceived security risks and effort—and emphasizing features that mitigate ambivalence—such as perceived enjoyment and usefulness—booking platforms can create a more engaging and trustworthy user experience. These strategies can help enhance user satisfaction and promote retention, positioning platforms for sustained success in a competitive market.

Security concerns contribute significantly to user ambivalence. Platforms can prioritize trust-building by implementing robust and transparent data protection measures. Visible mechanisms, such as two-factor authentication, encryption, and security certifications (e.g., GDPR compliance or "Trusted Travel Platform" badges), displayed prominently during the booking process, may reassure users and enhance their confidence. Clear and accessible data policies are equally important, as they help users understand how their information is collected, stored, and used. Simplified processes, such as intuitive password recovery and streamlined security checks, improve usability while maintaining strong security standards. These measures can help reduce security-related ambivalence and foster a seamless user experience.

The moderating role of perceived enjoyment highlights the importance of creating an emotionally engaging user experience. Gamified features, intuitive navigation, visually appealing designs, and immersive interfaces (Li et al., 2015) may evoke positive emotional responses and reduce ambivalence. AI-driven personalized recommendations streamline decision-making and enhance user satisfaction. Features such as rewards for bookings, interactive trip planners, and tailored offers can introduce enjoyable dimensions to the platform, strengthening emotional connections and minimizing conflicting evaluations.

Ambivalence evolves as users interact with platforms, experience updates, or adapt to new features. Platforms can sustain engagement by maintaining consistent communication about improvements, such as enhanced security measures or usability upgrades. Regular updates foster transparency and trust. Tools such as user surveys and AI-driven

sentiment analysis allow platforms to identify and address emerging concerns proactively. Personalized loyalty programs, milestone rewards (e.g., celebrating user anniversaries), and tailored offers strengthen emotional connections and reduce uncertainty. Strategies such as iterative updates aligned with user feedback and transparent communication about security changes support long-term engagement and minimize the risk of user disengagement.

Perceived usefulness plays a critical role in reducing ambivalence. Platforms should emphasize their practical benefits, such as price comparisons, loyalty programs, and exclusive deals, to strengthen their value proposition. Educational tools, such as interactive tutorials or guided walkthroughs, can help users feel more confident navigating advanced features. Real-time support through chatbots or live customer service during the booking process further enhances the user experience and addresses uncertainties. Features like clear refund policies, instant booking confirmations, and accessible FAQs can reinforce perceptions of platform reliability and usefulness.

Compliance with data privacy and security regulations, such as GDPR, is essential for addressing ambivalence related to security concerns. Aligning platform practices with legal standards ensures ethical and secure data management. Standardized certifications recognizing platforms for exceptional security and usability features can also serve as trust signals for users. Combined with clear and accessible privacy policies, these practices enhance user confidence and mitigate concerns about data misuse.

6.4. Limitations and future research

While this study provides valuable insights into the impact of ambivalence on user retention in online hotel booking platforms and its relationship with key antecedent beliefs, several limitations highlight opportunities for future research.

First, one limitation lies in the response options for the question, "How long have you been using hotel booking platforms?" The overlapping categories (e.g., "six months to one year" and "one year to three years") may have introduced ambiguity for respondents at category boundaries (e.g., exactly one year). To assess the potential impact, we conducted Monte Carlo simulations by probabilistically redistributing responses between overlapping categories. The results showed no meaningful effect on the overall outcomes, as key findings remained consistent across simulations. Although this variable was used only as a control and its impact is likely negligible, future research should adopt mutually exclusive categories (e.g., "six months to less than one year" and "one year to less than three years") to enhance clarity and reliability.

Second, the sample for this study was recruited from MTurk. While more diverse than traditional student samples, MTurk participants may not fully represent the broader population of online hotel booking users. Higher technological literacy and distinct motivations among MTurk users may influence responses related to perceived security risks, enjoyment, and ambivalence. Future studies should consider multi-channel recruitment strategies, such as partnering with online travel platforms to reach active users, leveraging social media advertisements, or conducting offline surveys at airports, travel expos, or hotels, to improve participant diversity and generalizability.

Third, while the sample size of 387 respondents meets established guidelines for PLS-SEM (Hair et al., 2022; Roy and Shaikh, 2024), it may limit the detection of smaller effect sizes or granular subgroup analyses (e.g., demographic comparisons). Larger samples could enable researchers to examine how ambivalence and retention vary across segments such as age or experience level. Combining MTurk sampling with platform-specific recruitment or consumer panels could enhance robustness and representation.

Fourth, the cross-sectional design limits this study's ability to infer causality or capture the temporal dynamics of ambivalence and retention. Ambivalence is inherently dynamic and may evolve as users interact with the platform over time. Future research should employ longitudinal designs to track ambivalence over time (e.g., at adoption, three months, six months) and assess its long-term effects on retention. Experimental studies could complement these designs by manipulating antecedents such as system transparency or interactivity to evaluate their causal impact on ambivalence and retention.

Fifth, this study focuses on online hotel booking platforms. While the findings provide significant insights within this context, generalizing them to other digital service platforms requires caution. Ambivalence's impact may differ in contexts such as e-commerce, social media, ride-hailing, or content streaming platforms, where user motivations and platform features vary. For example, hedonic motivations may amplify enjoyment's mitigating role in content streaming, while utilitarian drivers like trust and efficiency may dominate in e-commerce. Future research should replicate this study across different platforms to evaluate the broader applicability of the model.

Sixth, this study prioritizes four key antecedent beliefs—perceived security risk, effort, usefulness, and enjoyment— while excluding several other psychological constructs that could provide additional insights into ambivalence formation and outcomes. Variables such as trust (Castaldo et al., 2024; Park et al., 2012), habit (Turel, 2015), perceived control (Prestini and Sebastiani, 2021), attachment (Liu et al., 2020), decision-making styles (Klein and Sharma, 2022), and price fairness perception (Alderighi et al., 2022) may also play important roles. For example, perceived control may shape ambivalence by affecting users' sense of autonomy in platform interactions, trust could mitigate ambivalence by alleviating perceived risks, and habit might stabilize retention despite conflicting evaluations. Future research incorporating these constructs would yield a more comprehensive understanding of ambivalence and its broader implications.

Seventh, while this study highlights perceived security risk as a driver of ambivalence, it does not fully address the evolving landscape of data privacy and digital trust. Future research could explore how recent privacy regulations, such as GDPR, are associated with user perceptions of security over time. Longitudinal studies could examine whether heightened public awareness of privacy issues coincides with changes in ambivalence or greater trust in compliant platforms. Investigating the relationships among platform transparency, data protection practices, and perceived trustworthiness would provide a deeper understanding of security-related ambivalence.

Eighth, this study does not consider individual differences that may shape the relationship between ambivalence and retention. Factors such as personality traits, technology readiness, risk aversion, decision-making styles, and prior experience with technology may influence these dynamics. Future research could explore the potential moderating role of these factors to better capture variability in user behavior and ambivalence resolution processes.

Finally, while perceived enjoyment is identified as a key moderator in the ambivalence-retention relationship, other potential moderators and mediators have not been examined. Constructs such as habit, brand loyalty, trust, or platform familiarity may buffer the effects of ambivalence, while emotional processes such as anxiety, frustration, or anticipatory regret could provide additional insights into the mechanisms linking ambivalence and retention. Future research focusing on these constructs would contribute to a richer understanding of the complex interactions underlying ambivalence.

7. Conclusion

This study advances the understanding of ambivalence in IS contexts by demonstrating its critical role in post-adoption behaviors. By examining the interplay between key platform features—perceived usefulness, enjoyment, effort, and security risk—and ambivalence, this research offers insights into how cognitive and emotional conflicts are linked to user retention in competitive digital environments. The findings present a nuanced perspective on the coexistence of positive and

negative evaluations, emphasizing how ambivalence may challenge user commitment while highlighting the buffering role of perceived enjoyment.

The study makes significant theoretical contributions by moving beyond traditional satisfaction-focused models and positioning ambivalence as a pivotal construct in user behavior frameworks, aligning with the emerging bivariate unipolar perspective (Yu and Davis, 2024). Extending ambivalence research to online hotel booking platforms addresses a critical gap in the literature and broadens the understanding of post-adoption decision-making in digital service environments. These insights enrich IS theories by illustrating the multifaceted nature of user experience and the interplay of positive and negative beliefs.

From a practical perspective, this study offers actionable recommendations for platform designers, managers, and marketers. Strategies to address user ambivalence—such as improving security perceptions, minimizing perceived effort, and enhancing enjoyment and usefulness—are associated with fostering trust, engagement, and loyalty. Platforms that proactively address ambivalence through thoughtful design and transparent communication are better positioned to reduce churn, sustain user satisfaction, and maintain a competitive advantage in a dynamic and crowded marketplace.

CRediT authorship contribution statement

Yixiu Yu: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Fred Davis:** Writing – review & editing, Validation, Supervision, Funding acquisition, Data curation, Conceptualization. **Eric Walden:** Writing – review & editing, Validation, Supervision. **Ofir Turel:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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